Building and Configuring Basic Network Topologies using Cisco Packet Tracer

Aim

To build simple network topologies using Cisco Packet Tracer — ranging from a direct PC-to-PC link to a router-based topology connecting two separate LANs — and to configure appropriate IP addressing for each topology, then verify connectivity using the ping utility.

Theory

Cisco Packet Tracer is a cross-platform visual simulation tool designed by Cisco Systems that allows users to create network topologies and imitate the behaviour of modern computer networks. The software allows users to simulate the configuration of Cisco routers and switches using a simulated command line interface (CLI). Packet Tracer makes use of a drag-and-drop user interface, allowing users to add and remove simulated network devices as they see fit. The software is mainly intended for Cisco Networking Academy (CCNA) students as an educational tool for learning fundamental networking concepts.

A network can be built using different intermediary devices, and the choice of device affects how signals, collisions, and broadcasts are handled across the network:

- Direct (crossover) cable – connects two like devices (e.g., PC to PC) without any intermediary device.
- Hub – a Layer 1 device that simply regenerates and broadcasts the incoming signal out of all other ports. It creates a single collision domain and a single broadcast domain, and operates in half-duplex.
- Switch – a Layer 2 device that learns MAC addresses and forwards frames only to the relevant port, creating a separate collision domain per port while keeping a single broadcast domain.
- Repeater – a Layer 1 device used to regenerate and boost a signal so that it can travel beyond the standard cable-length limitation, without altering or filtering the data.
- Router – a Layer 3 device that connects two or more different networks (subnets) and forwards packets between them based on a routing table. When the networks are directly attached to the router's own interfaces, the router automatically adds a connected route for each one, so no manual configuration beyond the interface IP addresses is required for the two networks to reach each other.

In this manual, six progressively complex topologies are built and tested: (1) PC–PC, (2) PC–Hub–PC, (3) PC–Switch–PC, (4) PC–Switch–Switch–PC, (5) PC–Switch–Repeater–Switch–PC, and (6) PC–Switch–Router–Switch–PC. For each, appropriate IP addresses are configured on the end devices, cabling is completed, and connectivity is verified using the ping utility from the PC command prompt / simulation mode. All screenshots below are taken directly from the completed Packet Tracer implementation.

General Steps (applicable to every topology)

1. Open Cisco Packet Tracer and start a new blank workspace.
2. From the device panel, drag the required end devices (PCs/Laptops) and intermediary devices (Hub / Switch / Repeater / Router) onto the workspace as listed for that connection.

3. Use the Connections panel to cable the devices — use Copper Straight-Through cables between unlike devices (PC–Hub, PC–Switch, Switch–Router) and Copper Cross-Over cables between like devices (PC–PC, Switch–Switch), or simply use the Automatic cable option.
4. Click each PC/Laptop → Desktop tab → IP Configuration, and assign the Static IP Address, Subnet Mask, and Default Gateway (where applicable) as given in the IP Addressing Table.
5. Where a router is used, configure its interfaces via the router's CLI (Config or Terminal tab).
6. Verify Layer-1 connectivity — all connection indicator dots at the ends of the cables should turn green.
7. Test end-to-end connectivity using the ping command from the Command Prompt of the source device, or by sending a Simple PDU in Simulation mode.
8. Save the Packet Tracer file (.pkt) and capture the topology and verification screenshots for the report.

Connection 1: PC to PC

Aim

To connect two PCs directly with a cross-over cable, assign static IP addresses, and verify connectivity using the ping command.

Theory

Two PCs with identical network interface cards can be connected directly using a copper cross-over cable, without any intermediary networking device. This is the simplest possible network and forms a single collision domain and a single broadcast domain.

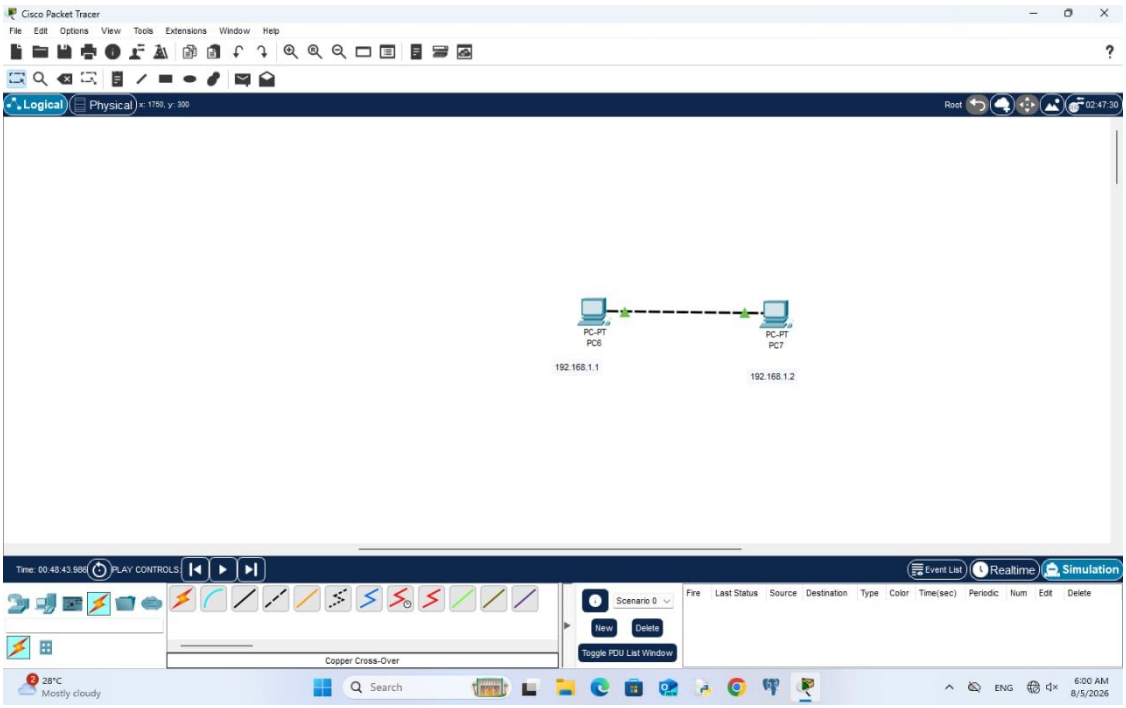
Since both PCs are on the same logical network, they must be assigned IP addresses from the same subnet so that they can communicate directly at Layer 3.

Devices Required

Device	Quantity
PC (End Device)	2
Copper Cross-Over Cable	1

Topology (as implemented in Packet Tracer)

Arrange and cable the devices in the logical workspace to form the sequence: PC to PC. The exact cabling order is given in the Procedure below.



IP Addressing Table

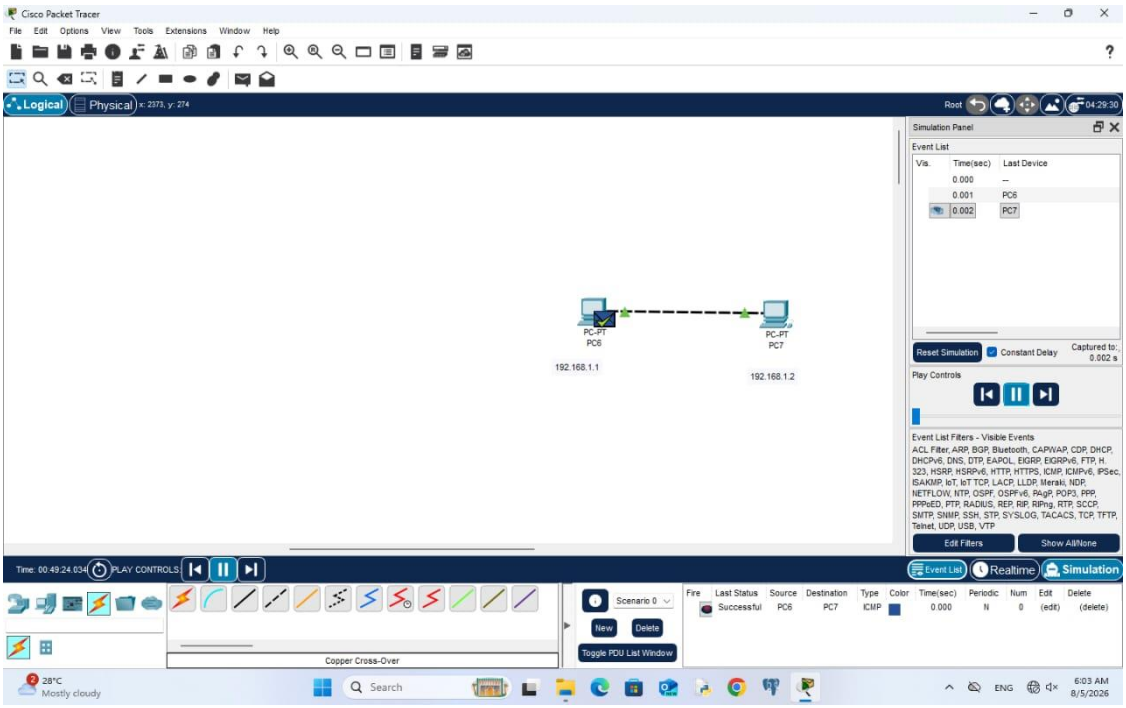
Device	Interface	IP Address	Subnet Mask	Default Gateway
PC6	FastEthernet0	192.168.1.1	255.255.255.0	—
PC7	FastEthernet0	192.168.1.2	255.255.255.0	—

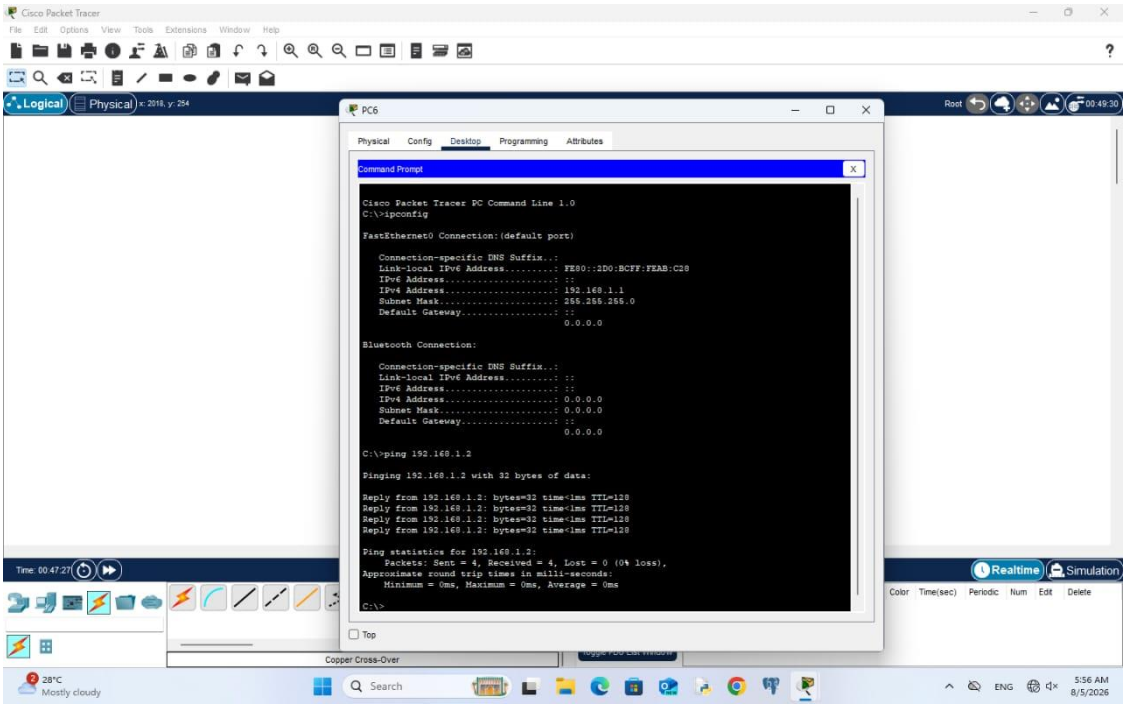
Procedure

1. Place two PC icons (PC6 and PC7) on the workspace.
2. Select the Copper Cross-Over cable from the Connections menu and connect PC6's FastEthernet0 port to PC7's FastEthernet0 port.
3. Click PC6 → Desktop → IP Configuration and set IP Address 192.168.1.1, Subnet Mask 255.255.255.0.
4. Click PC7 → Desktop → IP Configuration and set IP Address 192.168.1.2, Subnet Mask 255.255.255.0.
5. Wait for the link lights on both ends of the cable to turn green.
6. Open PC6 → Desktop → Command Prompt and run: ping 192.168.1.2

Result

The ping from PC6 to PC7 was successful (Reply from 192.168.1.2, 4 packets sent/4 received, 0% loss), confirming that a direct PC-to-PC connection with correctly configured static IP addresses provides full connectivity.


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical
Root
Simulation Panel
Event List
Vis. Time(sec) Last Device
0.000 --
0.001 PC6
0.002 PC7
PC-PT PC6
192.168.1.1
PC-PT PC7
192.168.1.2
Reset Simulation
Constant Delay
Captured to: 0.002 s
Play Controls
Event List Filters - Visible Events
ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoED, PTP, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP
Edit Filters
Show All/None
Time: 00:49:24.634
PLAY CONTROLS
Event List
Realtime
Simulation
Copper Cross-Over
Scenario 0
New
Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete
Successful PC6 PC7 ICMP 0.000 N 0 (edit) (delete)
28°C Mostly cloudy
Search
ENG
6:03 AM 8/5/2026


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical
Root
PC6
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig
FastEthernet0 Connection:(default port)
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: FE80::2D0:BCFF:FEAB:C28
IPv6 Address.....: ::
IPv4 Address.....: 192.168.1.1
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
0.0.0.0
Bluetooth Connection:
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
0.0.0.0
C:\>ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
Top
Time: 00:47:27
Realtime
Simulation
Copper Cross-Over
Color Time(sec) Periodic Num Edit Delete
28°C Mostly cloudy
Search
ENG
5:56 AM 8/5/2026

Connection 2: PC – Hub – PC

Aim

To connect PCs through an Ethernet hub, assign static IP addresses, and verify connectivity using the ping command.

Theory

A hub is a Layer 1 (Physical Layer) device that has no intelligence — it simply receives an electrical signal on one port and repeats/broadcasts it out of every other port. All devices connected to a hub share the same collision domain and the same broadcast domain, and communication is half-duplex.

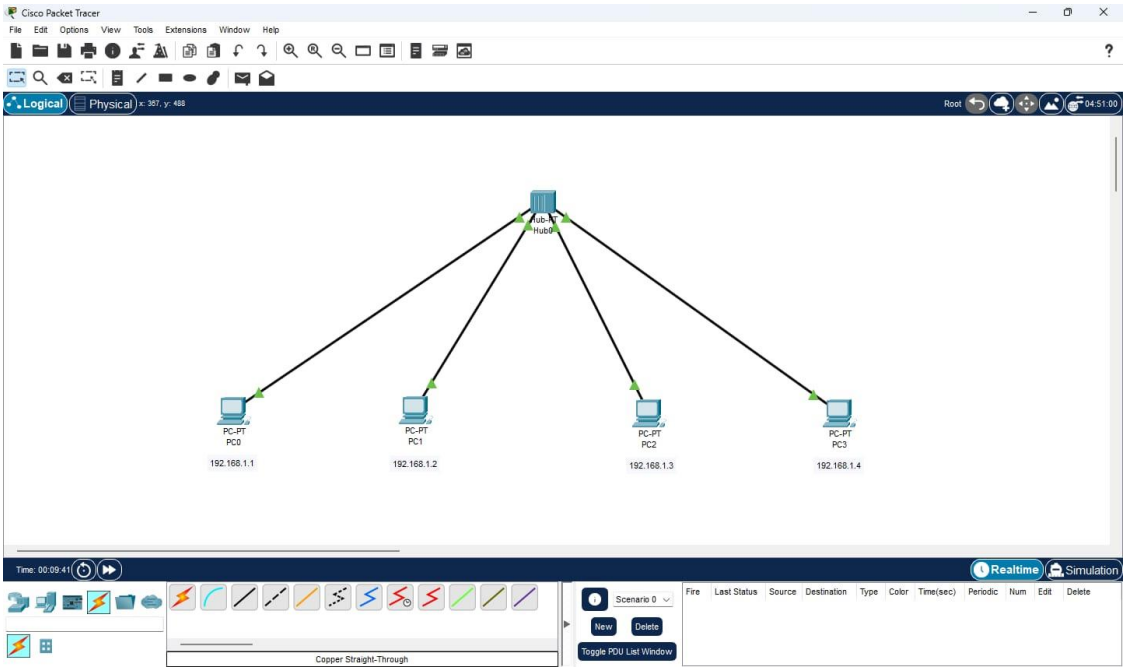
Because the hub does not perform any addressing or filtering, connectivity between the PCs depends entirely on their IP configuration being on the same subnet, exactly as in the direct-connection case.

Devices Required

Device	Quantity
PC (End Device)	4
Hub	1
Copper Straight-Through Cable	4

Topology (as implemented in Packet Tracer)

Arrange and cable the devices in the logical workspace to form the sequence: PC – Hub – PC. The exact cabling order is given in the Procedure below.



IP Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC0	FastEthernet0	192.168.1.1	255.255.255.0	—
PC1	FastEthernet0	192.168.1.2	255.255.255.0	—
PC2	FastEthernet0	192.168.1.3	255.255.255.0	—

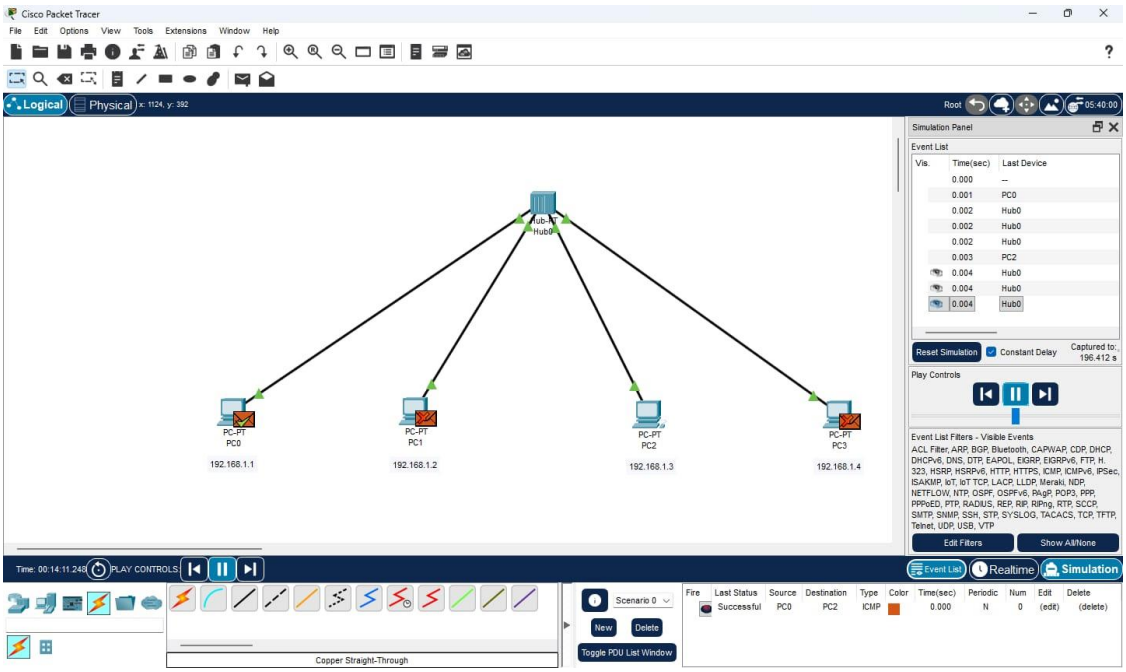
PC3	FastEthernet0	192.168.1.4	255.255.255.0	—

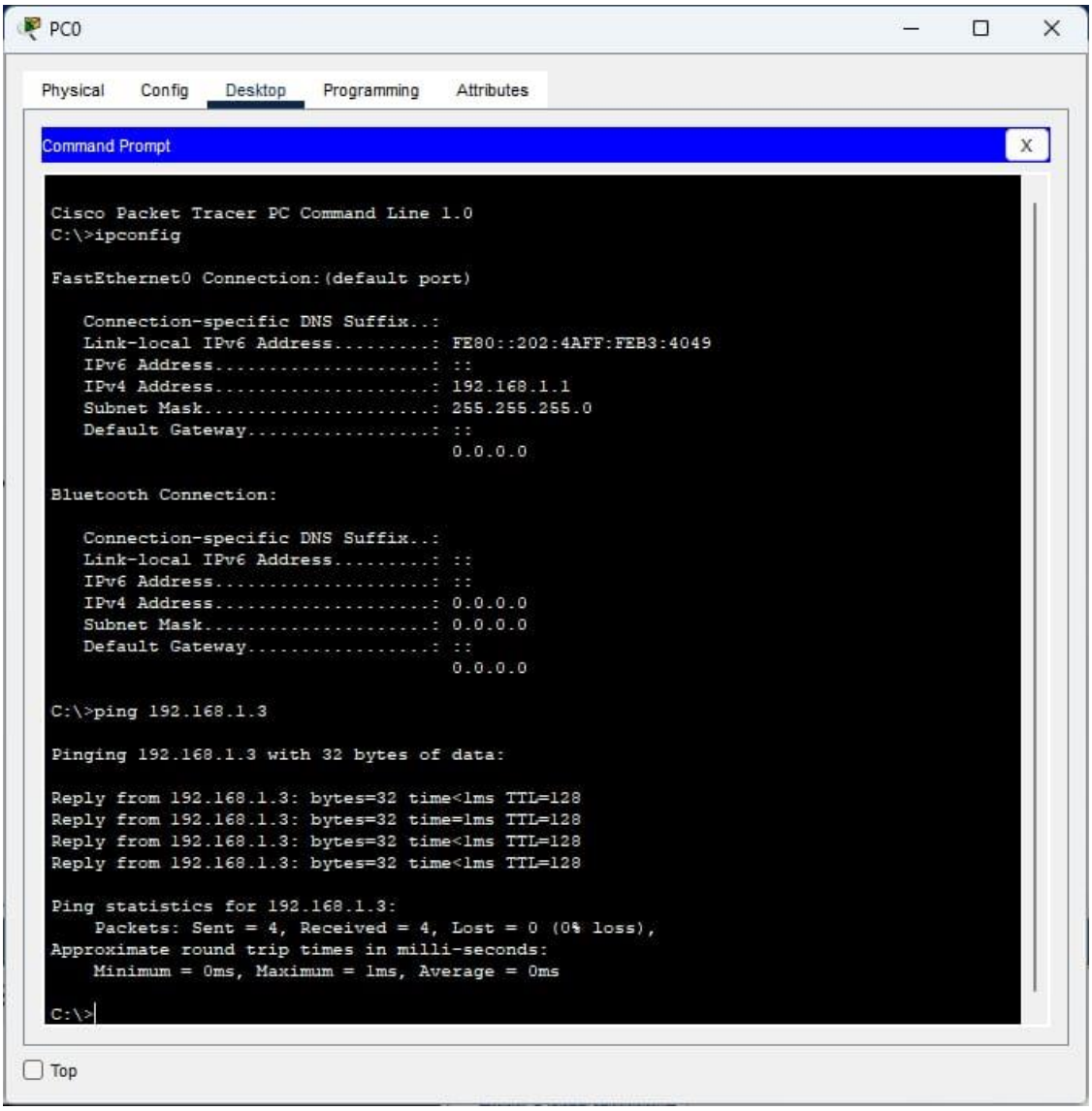
Procedure

1. Place four PC icons (PC0–PC3) and one Hub (Hub0) on the workspace.
2. Use Copper Straight-Through cables to connect the FastEthernet0 port of each of PC0, PC1, PC2, and PC3 to a separate port on Hub0.
3. Configure PC0's IP Address as 192.168.1.1, PC1's as 192.168.1.2, PC2's as 192.168.1.3, and PC3's as 192.168.1.4, all with Subnet Mask 255.255.255.0, via Desktop → IP Configuration.
4. Confirm that all link lights between the PCs and the hub are green.
5. From PC0's Command Prompt, run: ping 192.168.1.3 to verify connectivity through the hub.

Result

The ping from PC0 to PC2 (192.168.1.3) through the hub was successful (4 packets sent/4 received, 0% loss), confirming that the hub correctly repeats signals between all connected PCs on the same subnet, even though it operates purely at the physical layer with no intelligent forwarding.




PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig
FastEthernet0 Connection:(default port)
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: FE80::202:4AFF:FEB3:4049
IPv6 Address.....: ::
IPv4 Address.....: 192.168.1.1
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
0.0.0.0
Bluetooth Connection:
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
0.0.0.0
C:\>ping 192.168.1.3
Pinging 192.168.1.3 with 32 bytes of data:
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
Top

Connection 3: PC – Switch – PC

Aim

To connect PCs through a Layer 2 switch, assign static IP addresses, and verify connectivity using the ping command.

Theory

A switch is a Layer 2 (Data Link Layer) device that learns the MAC addresses of connected devices and builds a MAC address table, allowing it to forward frames intelligently only to the port where the destination device is located, instead of broadcasting to every port.

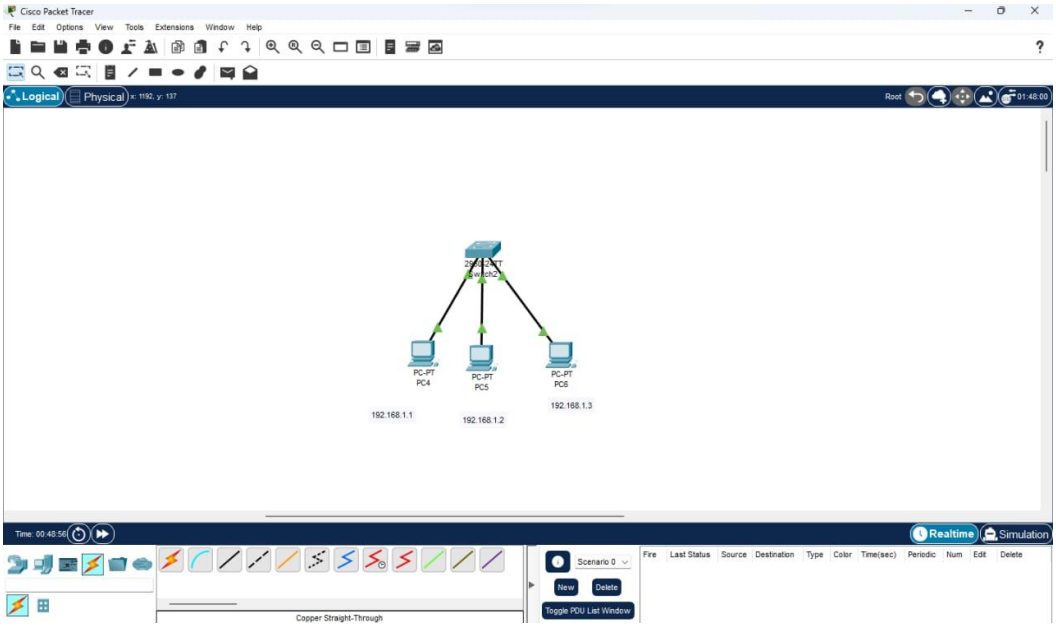
Each switch port is its own collision domain, which eliminates unnecessary collisions compared to a hub, while all ports still remain part of the same broadcast domain (in the absence of VLANs). As with the previous topology, all PCs must be configured on the same IP subnet.

Devices Required

Device	Quantity
PC (End Device)	3
Switch (2960 or similar)	1
Copper Straight-Through Cable	3

Topology (as implemented in Packet Tracer)

Arrange and cable the devices in the logical workspace to form the sequence: PC – Switch – PC. The exact cabling order is given in the Procedure below.



IP Addressing Table

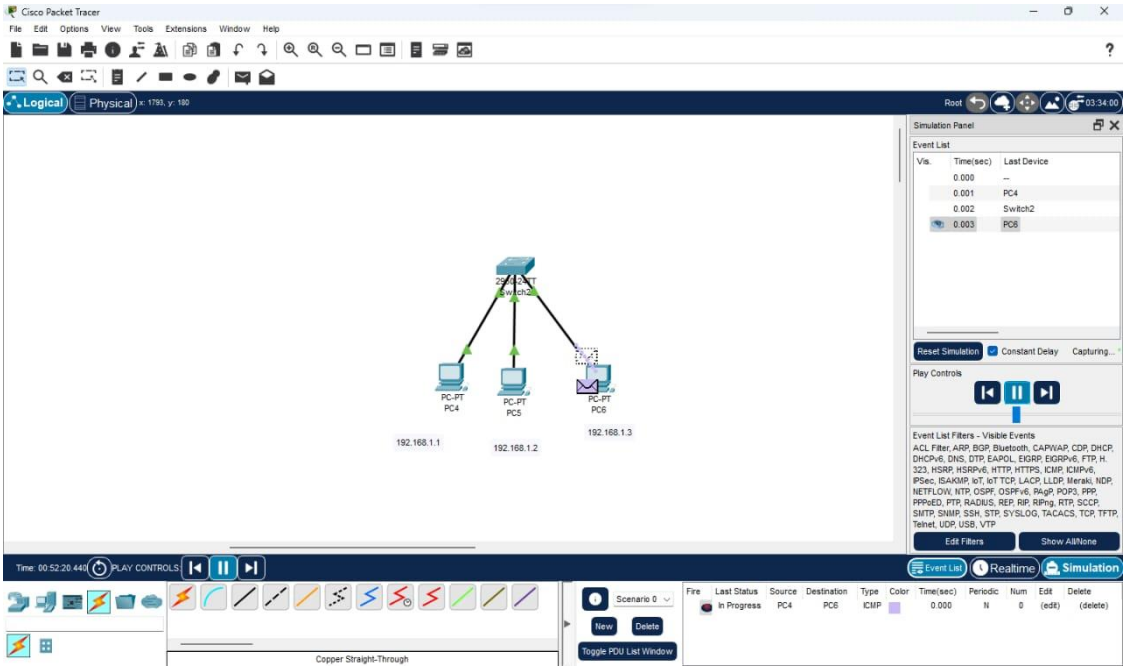
Device	Interface	IP Address	Subnet Mask	Default Gateway
PC4	FastEthernet0	192.168.1.1	255.255.255.0	—
PC5	FastEthernet0	192.168.1.2	255.255.255.0	—
PC6	FastEthernet0	192.168.1.3	255.255.255.0	—

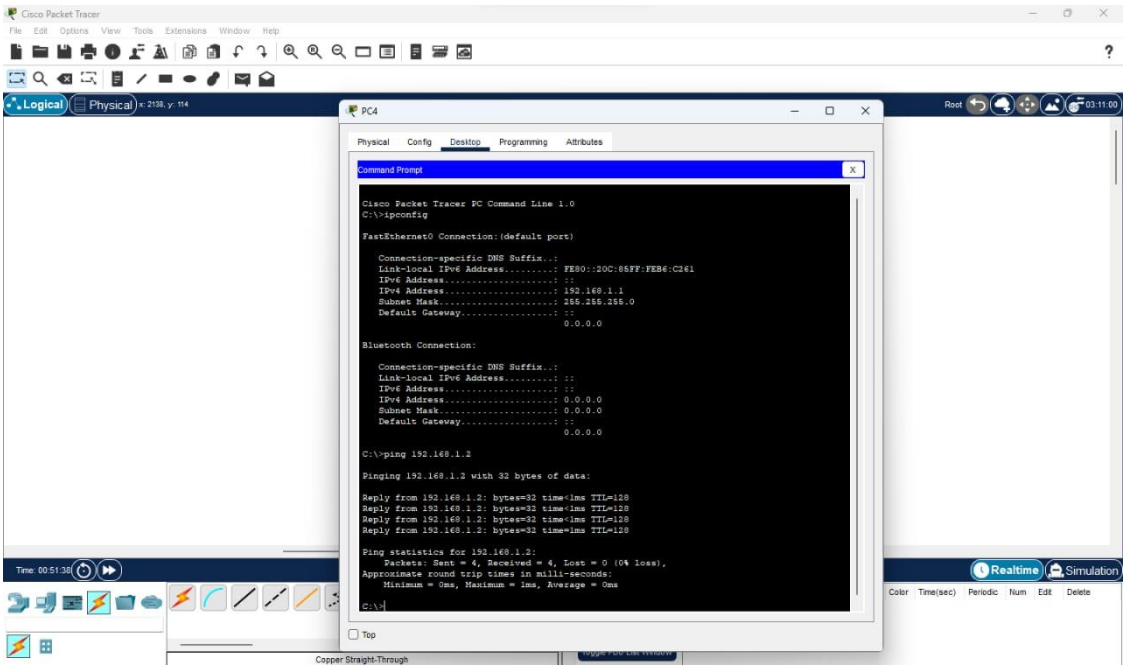
Procedure

1. Place three PC icons (PC4, PC5, PC6) and one Switch (Switch2) on the workspace.
2. Use Copper Straight-Through cables to connect PC4, PC5, and PC6 to three separate FastEthernet ports on Switch2.
3. Configure PC4's IP Address as 192.168.1.1, PC5's as 192.168.1.2, and PC6's as 192.168.1.3, all with Subnet Mask 255.255.255.0.
4. Wait for the switch port LEDs and cable end-points to turn green (this may take a few seconds while STP transitions the ports to forwarding state).
5. From PC4's Command Prompt, run: ping 192.168.1.2 to verify connectivity to PC5 across the switch, and repeat for PC6 (192.168.1.3).

Result

The pings from PC4 to the other PCs on Switch2 were successful (4 packets sent/4 received, 0% loss for PC4 → PC5), confirming correct connectivity. Using the switch instead of a hub also improves efficiency, since traffic is forwarded only to the intended port once the switch has learnt the MAC addresses of PC4, PC5, and PC6.





Connection 4: PC – Switch – Switch – PC

Aim

To connect two switched groups of end devices located on two cascaded switches, assign static IP addresses, and verify end-to-end connectivity.

Theory

Switches can be interconnected (cascaded) using an uplink port, which extends the reach of a single logical network across multiple physical switches. All ports across both switches still belong to the same broadcast domain and the same IP subnet, since no Layer 3 device separates them.

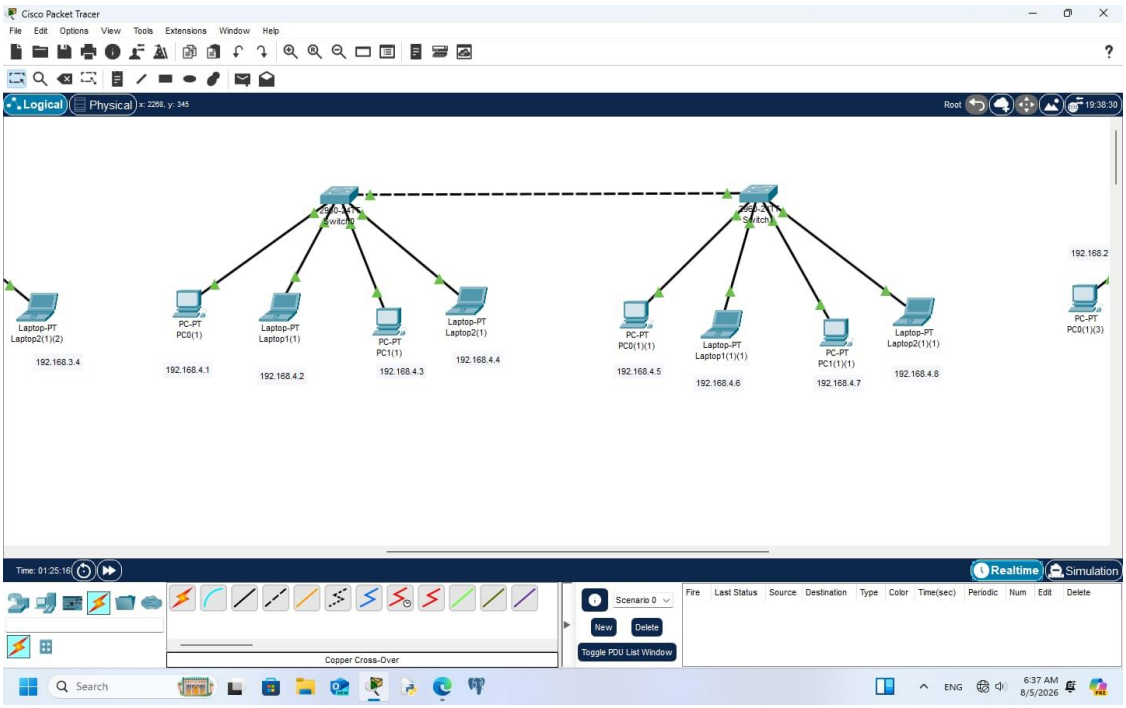
This topology demonstrates how a switched LAN can be scaled to accommodate more end devices — a mix of PCs and laptops — by adding more switches, while every device continues to reach any other device purely through Layer 2 forwarding.

Devices Required

Device	Quantity
PC (End Device)	4
Laptop (End Device)	4
Switch (2960 or similar)	2
Copper Straight-Through Cable	8
Copper Cross-Over Cable (Switch–Switch uplink)	1

Topology (as implemented in Packet Tracer)

Arrange and cable the devices in the logical workspace to form the sequence: PC – Switch – Switch – PC. The exact cabling order is given in the Procedure below.



IP Addressing Table

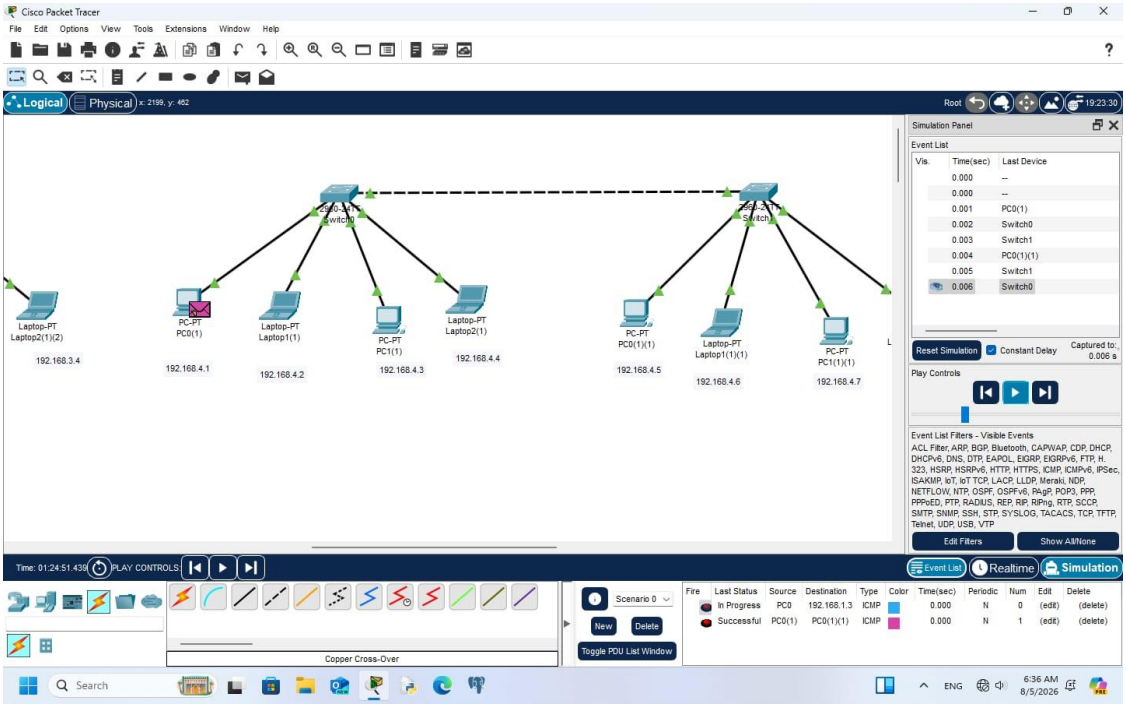
Device	Interface	IP Address	Subnet Mask	Default Gateway
PC0(1)	FastEthernet0	192.168.4.1	255.255.255.0	—
Laptop1(1)	FastEthernet0	192.168.4.2	255.255.255.0	—
PC1(1)	FastEthernet0	192.168.4.3	255.255.255.0	—
Laptop2(1)	FastEthernet0	192.168.4.4	255.255.255.0	—
PC0(1)(1)	FastEthernet0	192.168.4.5	255.255.255.0	—
Laptop1(1)(1)	FastEthernet0	192.168.4.6	255.255.255.0	—
PC1(1)(1)	FastEthernet0	192.168.4.7	255.255.255.0	—
Laptop2(1)(1)	FastEthernet0	192.168.4.8	255.255.255.0	—

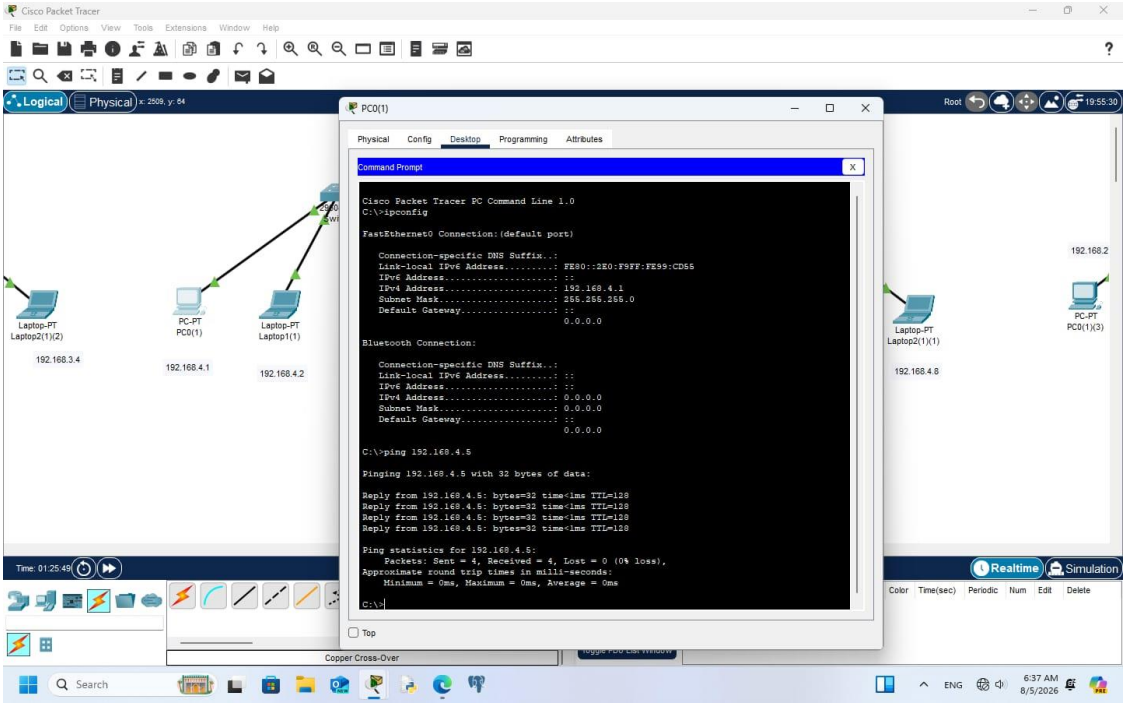
Procedure

1. Place four end devices (PC0(1), Laptop1(1), PC1(1), Laptop2(1)) and Switch0 on the left, and four end devices (PC0(1)(1), Laptop1(1)(1), PC1(1)(1), Laptop2(1)(1)) and Switch1 on the right.
2. Use Copper Straight-Through cables to connect each of the four left-hand devices to Switch0, and each of the four right-hand devices to Switch1.
3. Connect Switch0 to Switch1 using a Copper Cross-Over cable (or the Automatic cable, which selects the correct type).
4. Configure the left-hand devices with IP addresses 192.168.4.1 through 192.168.4.4 (Subnet Mask 255.255.255.0), in the order PC0(1), Laptop1(1), PC1(1), Laptop2(1).
5. Configure the right-hand devices with IP addresses 192.168.4.5 through 192.168.4.8 (Subnet Mask 255.255.255.0), in the order PC0(1)(1), Laptop1(1)(1), PC1(1)(1), Laptop2(1)(1).
6. Wait for all link indicators to turn green, then from PC0(1)'s Command Prompt run: ping 192.168.4.5

Result

The ping from PC0(1) (192.168.4.1) to PC0(1)(1) (192.168.4.5) across the two cascaded switches was successful (4 packets sent/4 received, 0% loss), confirming that the switch-to-switch uplink correctly forwards frames between the two segments of the same broadcast domain.





Connection 5: PC – Switch – Repeater – Switch – PC

Aim

To connect two switched groups of end devices through a repeater (to simulate distance extension) and verify end-to-end connectivity.

Theory

A repeater is a Layer 1 device that regenerates and re-times an incoming electrical/optical signal before forwarding it, allowing a network to be extended beyond the maximum cable-length limitation of a single segment (e.g., 100 m for copper Ethernet) without loss of signal quality.

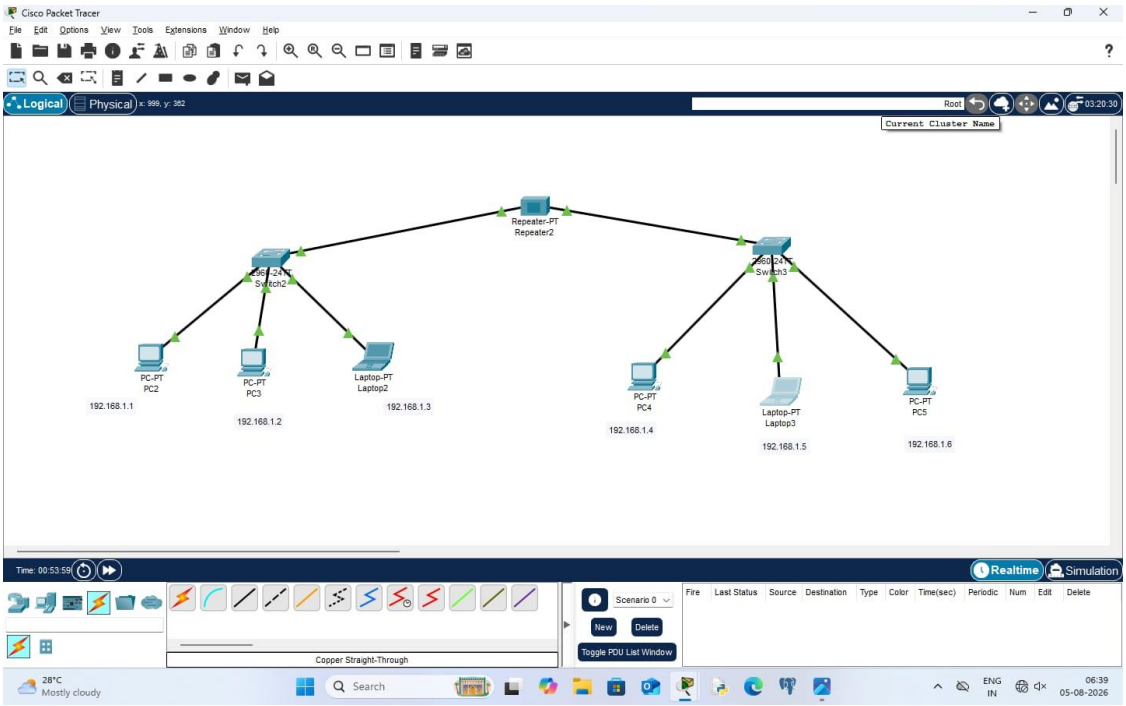
Placing a repeater between two switches does not create a new broadcast domain or subnet — it purely extends the physical segment between the switches at the physical layer, so all devices on both switches remain on the same IP subnet.

Devices Required

Device	Quantity
PC (End Device)	4
Laptop (End Device)	2
Switch (2960 or similar)	2
Repeater	1
Copper Straight-Through Cable	6

Topology (as implemented in Packet Tracer)

Arrange and cable the devices in the logical workspace to form the sequence: PC – Switch – Repeater – Switch – PC. The exact cabling order is given in the Procedure below.



IP Addressing Table

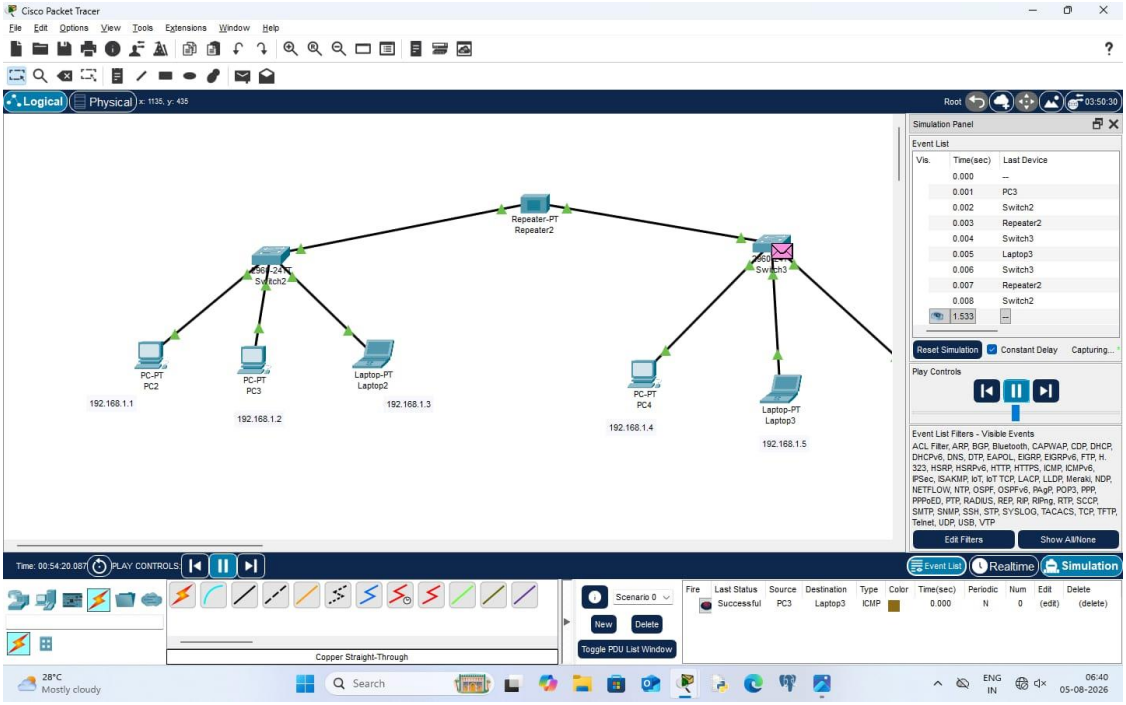
Device	Interface	IP Address	Subnet Mask	Default Gateway
PC2	FastEthernet0	192.168.1.1	255.255.255.0	—
PC3	FastEthernet0	192.168.1.2	255.255.255.0	—
Laptop2	FastEthernet0	192.168.1.3	255.255.255.0	—
PC4	FastEthernet0	192.168.1.4	255.255.255.0	—
Laptop3	FastEthernet0	192.168.1.5	255.255.255.0	—
PC5	FastEthernet0	192.168.1.6	255.255.255.0	—

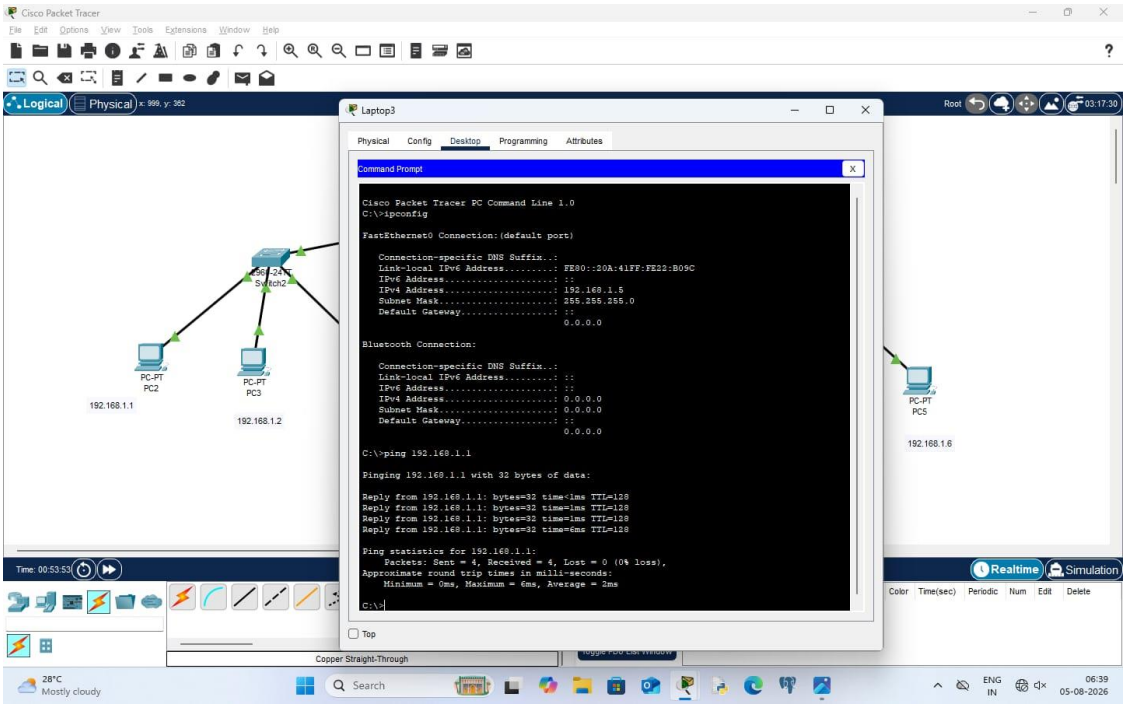
Procedure

1. Place PC2, PC3, and Laptop2 with Switch2 on the left, and PC4, Laptop3, and PC5 with Switch3 on the right, plus one Repeater (Repeater2) between the two switches.
2. Connect PC2, PC3, and Laptop2 to Switch2 using straight-through cables.
3. Connect PC4, Laptop3, and PC5 to Switch3 using straight-through cables.
4. Connect Switch2's uplink port to one port of Repeater2, and connect Repeater2's other port to Switch3's uplink port.
5. Configure PC2 = 192.168.1.1, PC3 = 192.168.1.2, Laptop2 = 192.168.1.3, PC4 = 192.168.1.4, Laptop3 = 192.168.1.5, and PC5 = 192.168.1.6, all with Subnet Mask 255.255.255.0.
6. Confirm all link lights are green, then from PC3's Command Prompt (or via a Simple PDU) run: ping 192.168.1.5 to test PC3 → Laptop3 across the repeater.
7. From Laptop3's Command Prompt, run: ping 192.168.1.1 to test Laptop3 → PC2 in the other direction.

Result

Both test pings were successful: PC3 → Laptop3 (192.168.1.5) completed with a 'Successful' status in the simulation event list, and Laptop3 → PC2 (192.168.1.1) returned 4 replies out of 4 packets sent (0% loss). This confirms that the repeater transparently regenerates the signal between the two switches without affecting Layer 2/3 connectivity.


Cisco Packet Tracer
Repeater-PT
Repeater2
Switch2
Switch3
PC-PT
PC2
PC3
PC4
Laptop-PT
Laptop2
Laptop3
192.168.1.1
192.168.1.2
192.168.1.3
192.168.1.4
192.168.1.5
Simulation Panel
Event List
Reset Simulation
Constant Delay
Play Controls
Edit Filters
Show All/None
Realtime
Simulation
Copper Straight-Through


Cisco Packet Tracer
Laptop3
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig
FastEthernet0 Connection:(default port)
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: FE80::20A:41FF:FE22:B09C
IPv6 Address.....: ::
IPv4 Address.....: 192.168.1.5
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
0.0.0.0
Bluetooth Connection:
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
0.0.0.0
C:\>ping 192.168.1.1
Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=6ms TTL=128
Ping statistics for 192.168.1.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 6ms, Average = 2ms
C:\>
Top
PC-PT
PC5
192.168.1.6
Realtime
Simulation
Copper Straight-Through

Connection 6: PC – Switch – Router – Switch – PC

Aim

To build a network with two separate LANs connected through a single router, configure IP addressing for each LAN, and verify that PCs on one LAN can communicate with PCs/laptops on the other LAN.

Theory

A router is a Layer 3 device that interconnects two or more different networks (subnets) and forwards packets between them. Unlike a switch, a router does not forward broadcasts between its interfaces, so each interface represents a separate broadcast domain / IP subnet.

In this topology, a single router (Router0, Cisco ISR4331) is used, with each of its two Gigabit Ethernet interfaces directly attached to a different LAN through its own switch. Because both networks terminate directly on the router's own interfaces, the router automatically installs a directly-connected route for each network in its routing table — no manual ip route (static route) entries are required for the two LANs to reach each other; only the interface IP addresses need to be configured.

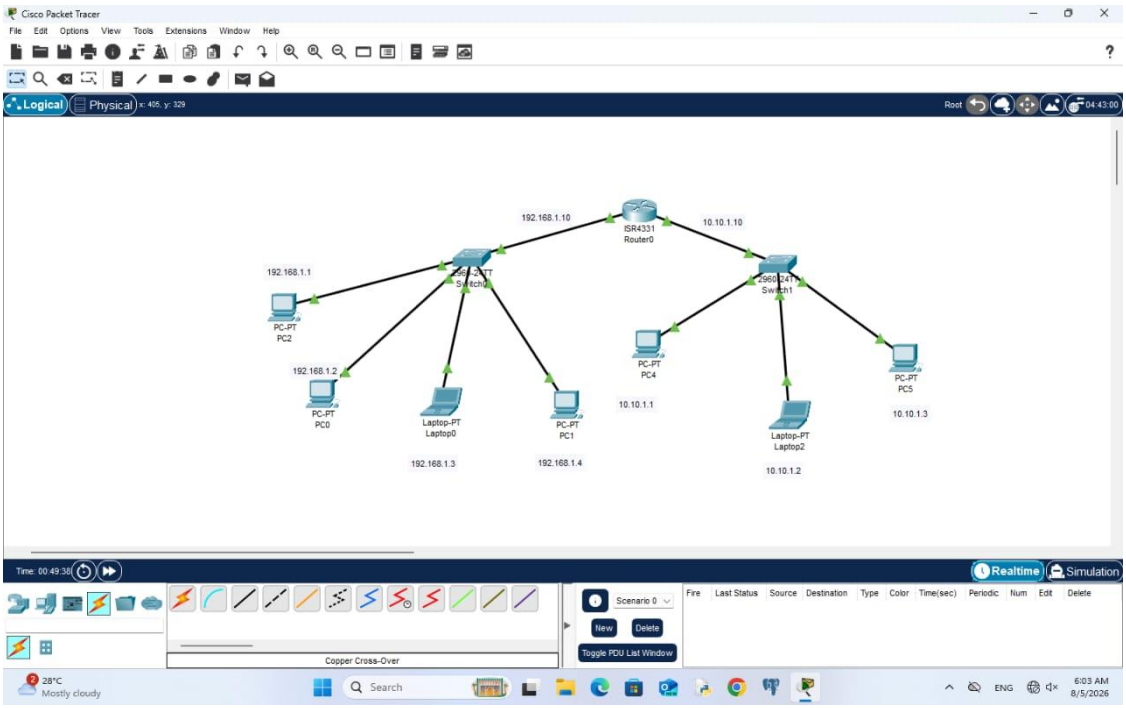
Seven end devices are used in total: four on LAN 1 (192.168.1.0/24, behind Switch0) and three on LAN 2 (10.10.1.0/24, behind Switch1). Each PC/laptop is configured with the router's interface address on its own LAN as its default gateway.

Devices Required

Device	Quantity
PC (End Device)	5
Laptop (End Device)	2
Switch (2960 or similar)	2
Router (ISR4331 or similar)	1
Copper Straight-Through Cable	9

Topology (as implemented in Packet Tracer)

Arrange and cable the devices in the logical workspace to form the sequence: PC – Switch – Router – Switch – PC. The exact cabling order is given in the Procedure below.



IP Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC2	FastEthernet0	192.168.1.1	255.255.255.0	192.168.1.10
PC0	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.10
Laptop0	FastEthernet0	192.168.1.3	255.255.255.0	192.168.1.10
PC1	FastEthernet0	192.168.1.4	255.255.255.0	192.168.1.10
Router0	GigabitEthernet0/0/0 (to Switch0)	192.168.1.10	255.255.255.0	—
Router0	GigabitEthernet0/0/1 (to Switch1)	10.10.1.10	255.255.255.0	—
PC4	FastEthernet0	10.10.1.1	255.255.255.0	10.10.1.10
Laptop2	FastEthernet0	10.10.1.2	255.255.255.0	10.10.1.10
PC5	FastEthernet0	10.10.1.3	255.255.255.0	10.10.1.10

Procedure

1. Place PC2, PC0, Laptop0, and PC1 with Switch0 on the LAN 1 side, and PC4, Laptop2, and PC5 with Switch1 on the LAN 2 side, with one Router (Router0, ISR4331) between the two switches.
2. Connect PC2, PC0, Laptop0, and PC1 to Switch0 using straight-through cables.
3. Connect PC4, Laptop2, and PC5 to Switch1 using straight-through cables.
4. Connect Switch0 to Router0's GigabitEthernet0/0/0 port, and Switch1 to Router0's GigabitEthernet0/0/1 port, using straight-through cables.
5. Configure the LAN 1 devices: PC2 = 192.168.1.1, PC0 = 192.168.1.2, Laptop0 = 192.168.1.3, PC1 = 192.168.1.4 (Subnet Mask 255.255.255.0, Gateway 192.168.1.10).
6. Configure the LAN 2 devices: PC4 = 10.10.1.1, Laptop2 = 10.10.1.2, PC5 = 10.10.1.3 (Subnet Mask 255.255.255.0, Gateway 10.10.1.10).
7. Open Router0's CLI (Config or Terminal tab) and configure GigabitEthernet0/0/0 with 192.168.1.10/24 and GigabitEthernet0/0/1 with 10.10.1.10/24, enabling both interfaces with no shutdown.

8. Wait for all connection indicators to turn green.
9. From PC1's Command Prompt (or via a Simple PDU), run ping 10.10.1.1 to verify that a device on LAN 1 can reach a device on LAN 2 through the router.
10. From PC0's Command Prompt, run ping 10.10.1.2 to further verify connectivity to Laptop2 on LAN 2.

Router / Switch Configuration Commands

Router0 configuration

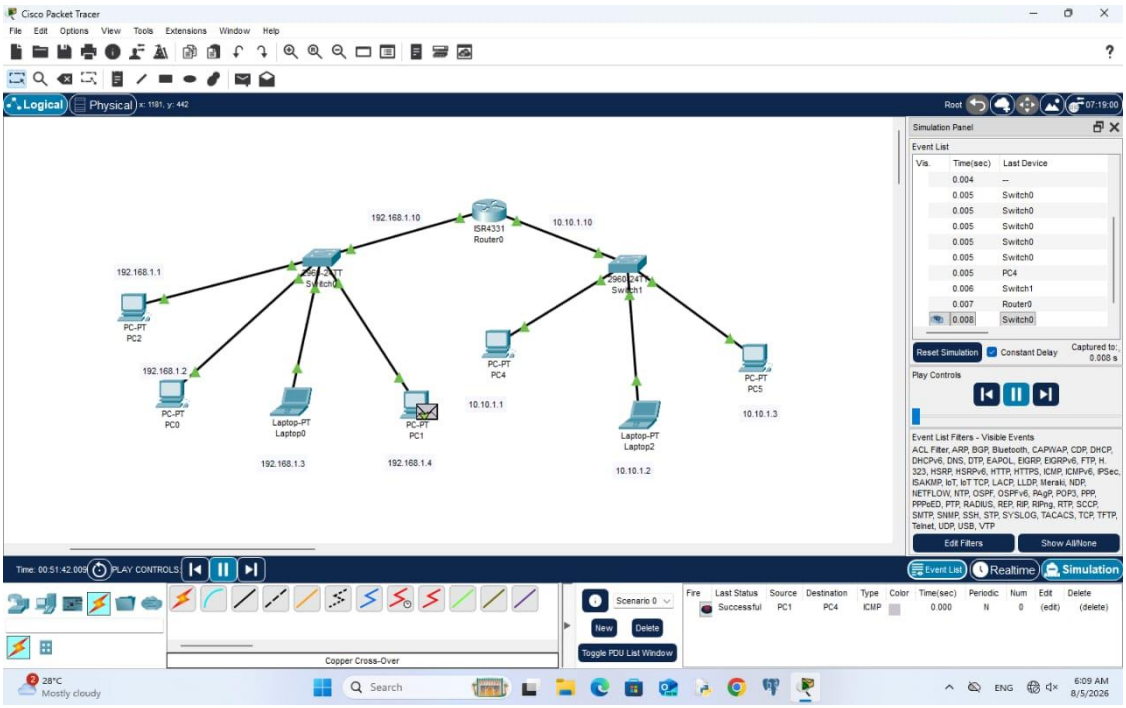
```
Router> enable
Router# configure terminal
Router(config)# hostname Router0
Router0(config)# interface gigabitEthernet 0/0/0
Router0(config-if)# ip address 192.168.1.10 255.255.255.0
Router0(config-if)# no shutdown
Router0(config-if)# exit
Router0(config)# interface gigabitEthernet 0/0/1
Router0(config-if)# ip address 10.10.1.10 255.255.255.0
Router0(config-if)# no shutdown
Router0(config-if)# exit
Router0(config)# exit
Router0# write memory
```

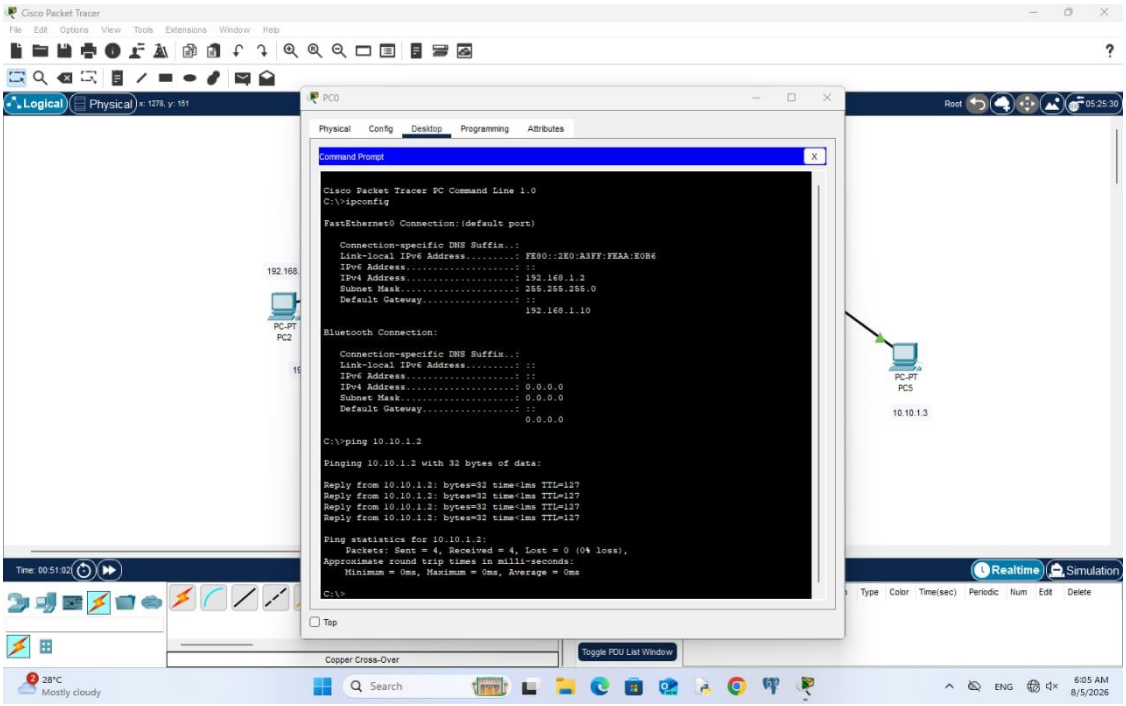
Verification commands

```
Router0# show ip route
Router0# show ip interface brief
Router0# ping 10.10.1.1
```

Result

The ping from PC1 (LAN 1, 192.168.1.4) to PC4 (LAN 2, 10.10.1.1) was successful, and the ping from PC0 (192.168.1.2) to Laptop2 (10.10.1.2) also completed with 4 replies out of 4 packets sent (0% loss, TTL reduced by one hop). This confirms that Router0's two directly-connected interfaces correctly route traffic between the 192.168.1.0/24 and 10.10.1.0/24 networks — the show ip route output on Router0 displays both networks as directly connected ('C') routes, which is sufficient for full connectivity between the two LANs in this single-router topology.


Cisco Packet Tracer
Logical
Physical
ISR4331
Router0
192.168.1.10
10.10.1.10
192.168.1.1
192.168.1.2
PC-PT
PC2
PC-PT
PC0
Laptop-PT
Laptop0
192.168.1.3
PC-PT
PC1
192.168.1.4
PC-PT
PC4
10.10.1.1
Laptop-PT
Laptop2
10.10.1.2
PC-PT
PC5
10.10.1.3
Simulation Panel
Event List
Reset Simulation
Constant Delay
Play Controls
Edit Filters
Show All/None
Realtime
Simulation
Copper Cross-Over
Toggle PDU List Window


Cisco Packet Tracer
PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig
FastEthernet0 Connection:(default port)
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: FE80::2E0:A3FF:FEAA:E0B6
IPv6 Address.....: ::
IPv4 Address.....: 192.168.1.2
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
192.168.1.10
Bluetooth Connection:
Connection-specific DNS Suffix..:
Link-local IPv6 Address.....: ::
IPv6 Address.....: ::
IPv4 Address.....: 0.0.0.0
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
0.0.0.0
C:\>ping 10.10.1.2
Pinging 10.10.1.2 with 32 bytes of data:
Reply from 10.10.1.2: bytes=32 time<1ms TTL=127
Reply from 10.10.1.2: bytes=32 time<1ms TTL=127
Reply from 10.10.1.2: bytes=32 time<1ms TTL=127
Reply from 10.10.1.2: bytes=32 time<1ms TTL=127
Ping statistics for 10.10.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
Top
PC-PT
PC2
PC-PT
PC5
10.10.1.3
Realtime
Simulation
Copper Cross-Over
Toggle PDU List Window

Conclusion

Six network topologies of increasing complexity — PC-to-PC, PC-Hub-PC, PC-Switch-PC, PC-Switch-Switch-PC, PC-Switch-Repeater-Switch-PC, and PC-Switch-Router-Switch-PC — were successfully built and tested in Cisco Packet Tracer. Appropriate static IP addressing was configured on all end devices, and connectivity was verified using the ping utility in each case, exactly as captured in the screenshots throughout this manual. The final topology additionally demonstrated routing between two separate LANs (192.168.1.0/24 and 10.10.1.0/24) through a single router, confirming that the router's directly-connected routes are sufficient to forward packets correctly between the two subnets.